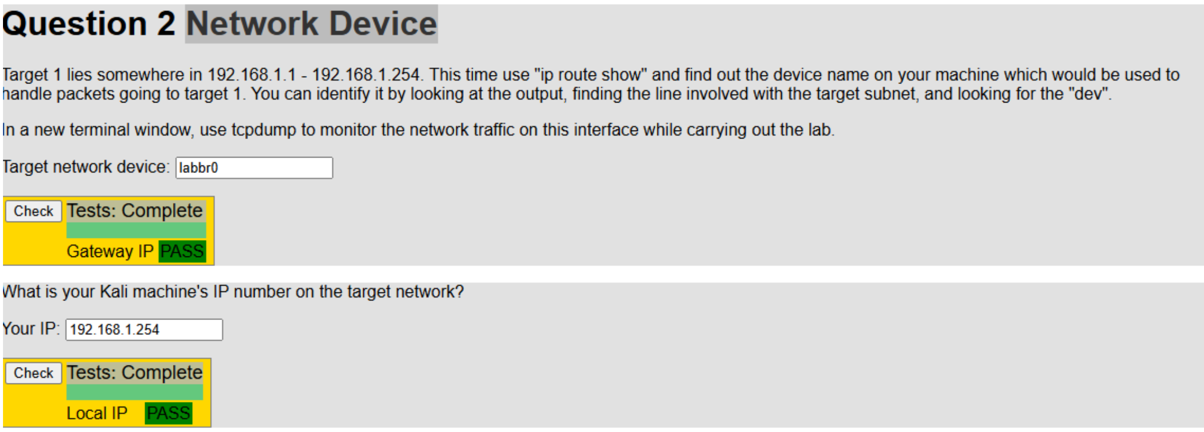
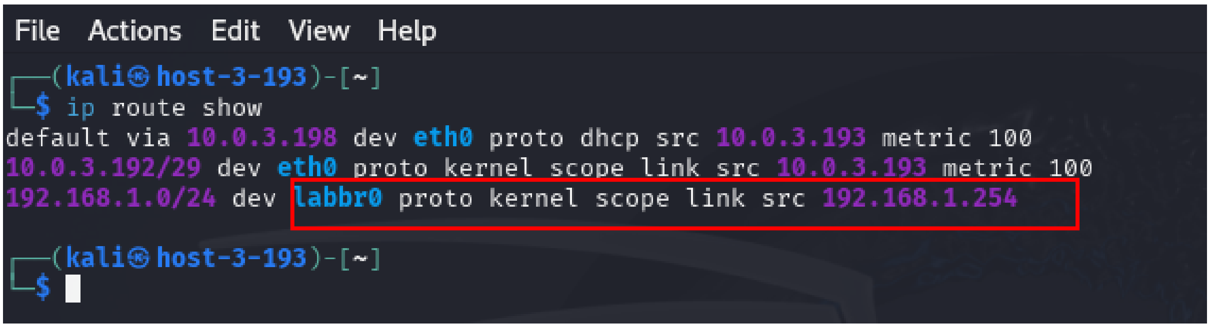
LAB 7 - Metasploit Exploit Framework (MSF) - Auxiliary Modules and Client-side Exploits

Q2. Network Device :

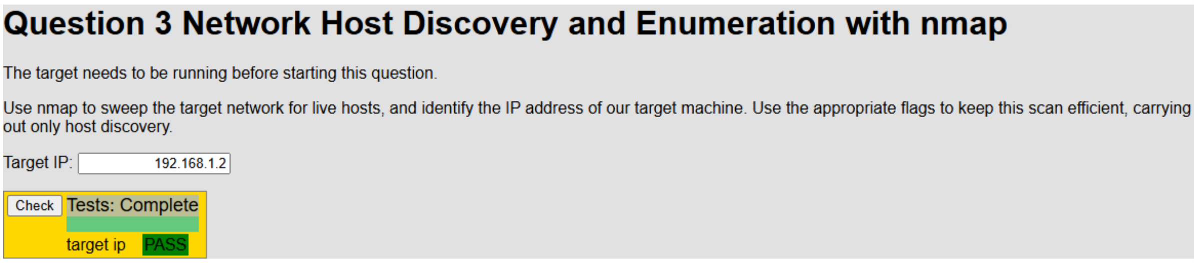
ip route show – to check "Your IP", Target network device.





Question 3 - Network Host Discovery and Enumeration with nmap

- **nmap -n -O to get the details pof target IP and port details.**



-

```
┌──(kali㉿host-3-193)-[~]
└─$ nmap -n -O 192.168.1.1-254
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 11:28 GMT
Nmap scan report for 192.168.1.2
Host is up (0.00043s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 00:0C:B1:00:07:90 (Salland Engineering (Europe) BV)
Device type: general purpose
Running: Microsoft Windows XP|2003
OS CPE: cpe:/o:microsoft:windows_xp::sp3:embedded cpe:/o:microsoft:windows_xp::sp2 cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_s
erver_2003
OS details: Microsoft Windows XP SP2 or SP3, or Windows Embedded Standard 2009, Microsoft Windows XP SP2 or SP3, or Windows Server 2003
Network Distance: 1 hop

Nmap scan report for 192.168.1.254
Host is up (0.000073s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
53/tcp open  domain
Device type: general purpose
Running: Linux 2.6.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 2.6.32, Linux 5.0 - 6.2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 254 IP addresses (2 hosts up) scanned in 5.15 seconds
```

To perform nmap portscan – nmap -sS “TARGET IP” -oX to save file in xml format

Port Scanning for Target Services

Port scan the target, first performing a nmap portscan on the same subnet, for all default TCP ports, with no name resolution, no host discovery, and the use the â□□oX flag to save the details of the scan to a file. Save the file to /home/kali/subnet1

How many TCP ports are open?

3

What is the format of the saved scan results file?

xml

```
┌──(kali㉿host-5-105)-[~]
└─$ nmap -sS 192.168.1.2 -oX subnet1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-19 11:56 GMT
Nmap scan report for 192.168.1.2
Host is up (0.020s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 00:0C:B1:00:07:90 (Salland Engineering (Europe) BV)

Nmap done: 1 IP address (1 host up) scanned in 0.63 seconds
```

Figure: Port Scan of Target

```
┌──(kali㉿host-7-209)-[~]
└─$ cat subnet1
```

```
┌──(kali㉿host-3-193)-[~]
└─$ nmap -sS 192.168.1.2 -oX subnet1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 11:37 GMT
Nmap scan report for 192.168.1.2
Host is up (0.012s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 00:0C:B1:00:07:90 (Salland Engineering (Europe) BV)

Nmap done: 1 IP address (1 host up) scanned in 0.74 seconds

┌──(kali㉿host-3-193)-[~]
└─$ cat subnet1
```


```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE nmaprun>
<?xml-stylesheet href="file:///usr/share/nmap/nmap.xsl" type="text/xsl"?>
<!-- Nmap 7.95 scan initiated Wed Mar 18 11:37:52 2026 as: /usr/lib/nmap/nmap -&#45;privileged -sS -oX subnet1 192.168.1.2 -->
<nmaprun scanner="nmap" args="/usr/lib/nmap/nmap -&#45;privileged -sS -oX subnet1 192.168.1.2" start="1773833872" startstr="Wed Mar 18 11:37:5
2 2026" version="7.95" xmloutputversion="1.05">
<scaninfo type="syn" protocol="tcp" numservices="1000" services="1,3-4,6-7,9,13,17,19-26,30,32-33,37,42-43,49,53,70,79-85,88-90,99-100,106,109
-111,113,119,125,135,139,143-144,146,161,163,179,199,211-212,222,254-256,259,264,280,301,306,311,340,366,389,406-407,416-417,425,427,443-445,4
58,464-465,481,497,500,512-515,524,541,543-545,548,554-555,563,587,593,616-617,625,631,636,646,648,666-668,683,687,691,700,705,711,714,720,722
,726,749,765,777,783,787,800-801,808,843,873,880,888,898,900-903,911-912,981,987,990,992-993,995,999-1002,1007,1009-1011,1021-1100,1102,1104-1
108,1110-1114,1117,1119,1121-1124,1126,1130-1132,1137-1138,1141,1145,1147-1149,1151-1152,1154,1163-1166,1169,1174-1175,1183,1185-1187,1192,119
8-1199,1201,1213,1216-1218,1233-1234,1236,1244,1247-1248,1259,1271-1272,1277,1287,1296,1300-1301,1309-1311,1322,1328,1334,1352,1417,1433-1434,
1443,1455,1461,1494,1500-1501,1503,1521,1524,1533,1556,1580,1583,1594,1600,1641,1658,1666,1687-1688,1700,1717-1721,1723,1755,1761,1782-1783,18
01,1805,1812,1839-1840,1862-1864,1875,1900,1914,1935,1947,1971-1972,1974,1984,1998-2010,2013,2020-2022,2030,2033-2035,2038,2040-2043,2045-2049
,2065,2068,2099-2100,2103,2105-2107,2111,2119,2121,2126,2135,2144,2160-2161,2170,2179,2190-2191,2196,2200,2222,2251,2260,2288,2301,2323,2366,2
381-2383,2393-2394,2399,2401,2492,2500,2522,2525,2557,2601-2602,2604-2605,2607-2608,2638,2701-2702,2710,2717-2718,2725,2800,2809,2811,2869,287
5,2909-2910,2920,2967-2968,2998,3000-3001,3003,3005-3006,3011,3017,3030-3031,3052,3071,3077,3128,3168,3211,3221,3260-3261,3268-3269,3283,3300-
3301,3306,3322-3325,3333,3351,3367,3369-3372,3389-3390,3404,3476,3493,3517,3527,3546,3551,3580,3659,3689-3690,3703,3737,3766,3784,3800-3801,38
09,3814,3826-3828,3851,3869,3871,3878,3880,3889,3905,3914,3918,3920,3945,3971,3986,3995,3998,4000-4006,4045,4111,4125-4126,4129,4224,4242,4279
,4321,4343,4443-4446,4449,4550,4567,4662,4848,4899-4900,4998,5000-5004,5009,5030,5033,5050-5051,5054,5060-5061,5080,5087,5100-5102,5120,5190,5
200,5214,5221-5222,5225-5226,5269,5280,5298,5357,5405,5414,5431-5432,5440,5500,5510,5544,5550,5555,5560,5566,5631,5633,5666,5678-5679,5718,573
```

Question 4 Metasploit Framework

Metasploit Database- Before performing the complex penetration tests, firstly check the status of msf database.

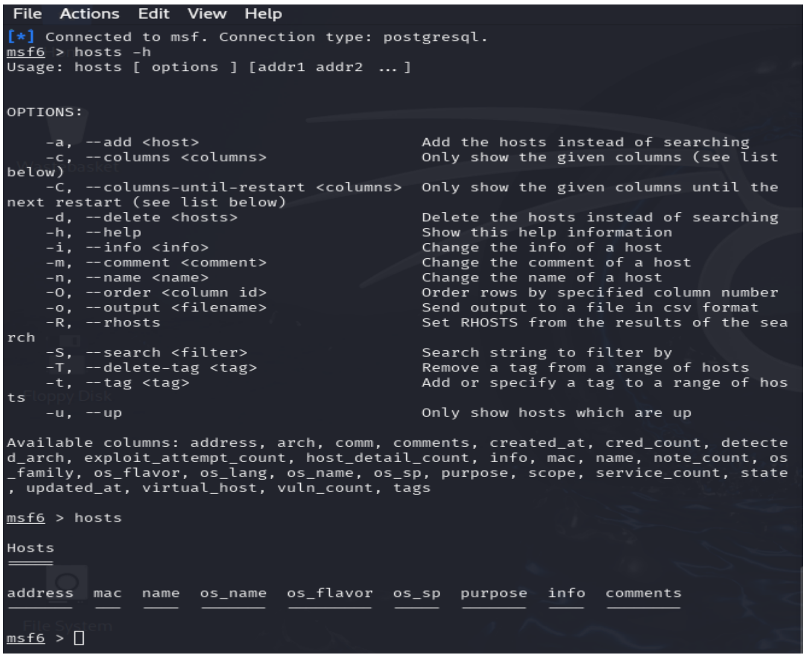
- Sudo msfdb status
- db_status

Then, check the target host in the database.

-hosts -h → Help

Now, check currently stored target host

-hosts

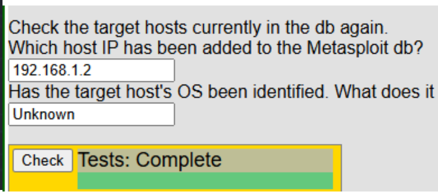


```
File  Actions  Edit  View  Help
[*] Connected to msf. Connection type: postgresql.
msf6 > hosts -h
Usage: hosts [ options ] [addr1 addr2 ... ]

OPTIONS:

    -a, --add <host>                      Add the hosts instead of searching
    -c, --columns <columns>               Only show the given columns (see list
below)
    -C, --columns-until-restart <columns>  Only show the given columns until the
next restart (see list below)
    -d, --delete <hosts>                  Delete the hosts instead of searching
    -h, --help                            Show this help information
    -i, --info <info>                     Change the info of a host
    -m, --comment <comment>               Change the comment of a host
    -n, --name <name>                     Change the name of a host
    -O, --order <column id>               Order rows by specified column number
    -o, --output <filename>               Send output to a file in csv format
    -R, --rhosts                          Set RHOSTS from the results of the sea
rch
    -S, --search <filter>                 Search string to filter by
    -T, --delete-tag <tag>                Remove a tag from a range of hosts
    -t, --tag <tag>                       Add or specify a tag to a range of hos
ts
    -u, --up                              Only show hosts which are up

Available columns: address, arch, comm, comments, created_at, cred_count, detecte
d_arch, exploit_attempt_count, host_detail_count, info, mac, name, note_count, os
_family, os_flavor, os_lang, os_name, os_sp, purpose, scope, service_count, state
, updated_at, virtual_host, vuln_count, tags

msf6 > hosts

Hosts
=====

address  mac  name  os_name  os_flavor  os_sp  purpose  info  comments
-------  ---  ----  -------  ---------  -----  -------  ----  --------

msf6 >
```

db_import – command help to get output from the other tools.

Now, After importing from other tools, checked the host again and got the details of IP Host and OS name.

```
msf6 > db_import subnet1
[*] Importing 'Nmap XML' data
[*] Import: Parsing with 'Nokogiri v1.13.10'
[*] Importing host 192.168.1.2
[*] Successfully imported /home/kali/subnet1
msf6 > hosts

Hosts
=====

address   mac        name  os_name  os_flavor  os_sp  purpose  info  comments
-------   ---        ----  -------  ---------  -----  -------  ----  --------
192.168.  00:0c:b1:        Unknown                    device
1.2       00:07:90
```



Question 5 Network Sweeping, Target Enumeration with Port Scanning and OS Fingerprinting

- To begin with network sweeping, target enumeration with port scanning and OS –
- **db_nmap** command will perform network scans
- **db_nmap -n -O –osscan-guess "Target Ips"** – Host discovery scan with OS and post scan.

```
msf6 > db_nmap -n -O --osscan-guess 192.168.1.0/24
[*] Nmap: Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 12:36 GMT
[*] Nmap: Nmap scan report for 192.168.1.2
[*] Nmap: Host is up (0.0033s latency).
[*] Nmap: Not shown: 997 closed tcp ports (reset)
[*] Nmap: PORT     STATE SERVICE
[*] Nmap: 135/tcp open  msrpc
[*] Nmap: 139/tcp open  netbios-ssn
[*] Nmap: 445/tcp open  microsoft-ds
[*] Nmap: MAC Address: 00:0C:B1:00:07:90 (Salland Engineering (Europe) BV)
[*] Nmap: Device type: general purpose
[*] Nmap: Running: Microsoft Windows XP|2003
[*] Nmap: OS CPE: cpe:/o:microsoft:windows_xp::sp3:embedded cpe:/o:microsoft:wind
ows_xp::sp2 cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_server_2003
[*] Nmap: OS details: Microsoft Windows XP SP2 or SP3, or Windows Embedded Standa
rd 2009, Microsoft Windows XP SP2 or SP3, or Windows Server 2003
[*] Nmap: Network Distance: 1 hop
[*] Nmap: Nmap scan report for 192.168.1.254
[*] Nmap: Host is up (0.000066s latency).
[*] Nmap: Not shown: 998 closed tcp ports (reset)
[*] Nmap: PORT   STATE SERVICE
[*] Nmap: 22/tcp open  ssh
[*] Nmap: 53/tcp open  domain
[*] Nmap: Device type: general purpose
[*] Nmap: Running: Linux 2.6.X|5.X
[*] Nmap: OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cp
e:/o:linux:linux_kernel:6
[*] Nmap: OS details: Linux 2.6.32, Linux 5.0 - 6.2
[*] Nmap: Network Distance: 0 hops
[*] Nmap: OS detection performed. Please report any incorrect results at https://
nmap.org/submit/ .
[*] Nmap: Nmap done: 256 IP addresses (2 hosts up) scanned in 5.07 seconds
msf6 >
```

Question 5 Network Sweeping, Target Enumeration with Port Scanning and OS Fingerprinting

From the msfconsole shell we can use nmap directly to perform network scans, and save the output findings into the MSF db, using the **db_nmap** command.

`db_nmap -n -O other nmap options`

Perform a Host Discovery scan on the entire target subnet, and a default Port Scan with OS Fingerprinting for any live hosts found, with no name resolution.

Which version of nmap did this command use?

7.95

What "Running:" does the scan suggest? Enter exactly.

Microsoft Windows XP|2003

Check Tests: Complete

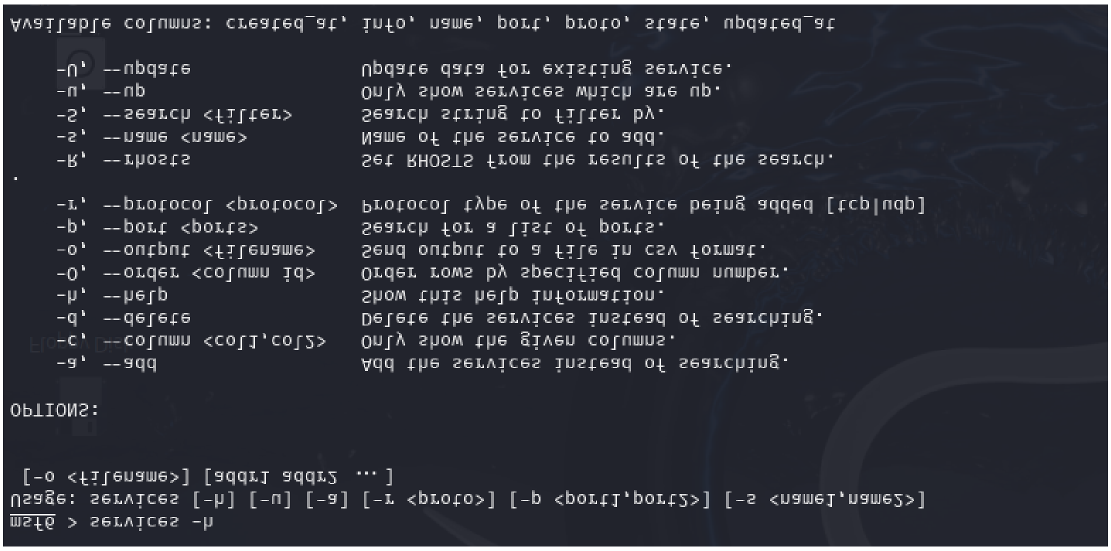
db_nmap complete	PASS
nmap version	PASS

Now, after host discovery with db_nmap, check hosts details again

-hosts

- To identify the services in the target host

Services -h

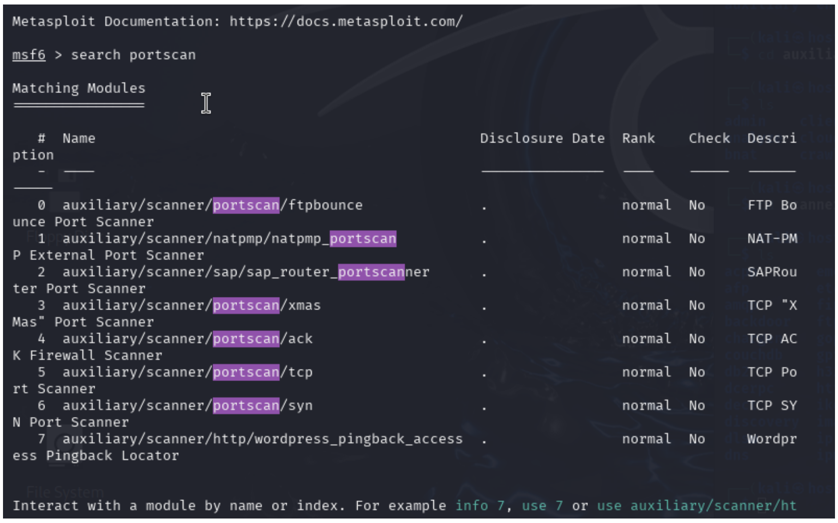


```
msf6 > services -h
Usage: services [-h] [-u] [-a] [-r <proto>] [-p <port1,port2>] [-s <name1,name2>]
 [-o <filename>] [addr1 addr2 ...]

OPTIONS:

    -a, --add                  Add the services instead of searching.
    -c, --column <col1,col2>   Only show the given columns.
    -d, --delete               Delete the services instead of searching.
    -h, --help                 Show this help information.
    -O, --order <column id>    Order rows by specified column number.
    -o, --output <filename>    Send output to a file in csv format.
    -p, --port <ports>         Search for a list of ports.
    -r, --protocol <protocol>  Protocol type of the service being added [tcp|udp]
.
    -R, --rhosts               Set RHOSTS from the results of the search.
    -s, --name <name>          Name of the service to add.
    -S, --search <filter>      Search string to filter by.
    -u, --up                   Only show services which are up.
    -U, --update               Update data for existing service.

Available columns: created_at, info, name, port, proto, state, updated_at
```

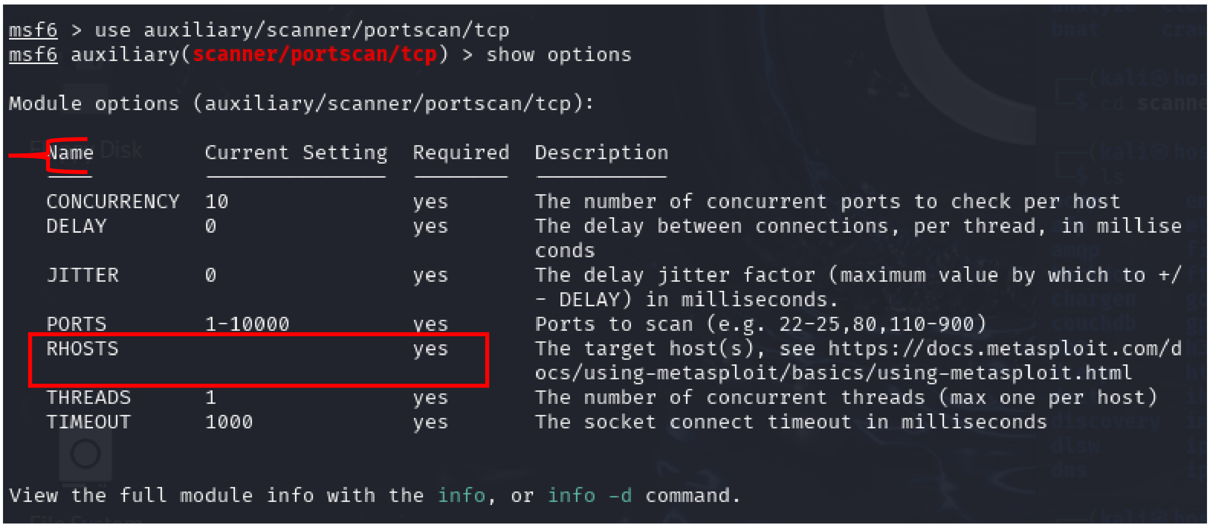
Question 6 - Auxiliary Modules

To check the list all of the auxiliary modules - **search type:auxiliary**

Now, find port scanner auxiliary scripts: - **search ports**



```
Metasploit Documentation: https://docs.metasploit.com/

msf6 > search portscan

Matching Modules
================

   #  Name                                              Disclosure Date  Rank    Check  Descri
ption
   -  ----                                              ---------------  ----    -----  ------
----
   0  auxiliary/scanner/portscan/ftpbounce              .                normal  No     FTP Bo
unce Port Scanner
   1  auxiliary/scanner/natpmp/natpmp_portscan          .                normal  No     NAT-PM
P External Port Scanner
   2  auxiliary/scanner/sap/sap_router_portscanner      .                normal  No     SAPRou
ter Port Scanner
   3  auxiliary/scanner/portscan/xmas                   .                normal  No     TCP "X
Mas" Port Scanner
   4  auxiliary/scanner/portscan/ack                    .                normal  No     TCP AC
K Firewall Scanner
   5  auxiliary/scanner/portscan/tcp                    .                normal  No     TCP Po
rt Scanner
   6  auxiliary/scanner/portscan/syn                    .                normal  No     TCP SY
N Port Scanner
   7  auxiliary/scanner/http/wordpress_pingback_access  .                normal  No     Wordpr
ess Pingback Locator

Interact with a module by name or index. For example info 7, use 7 or use auxiliary/scanner/ht
```

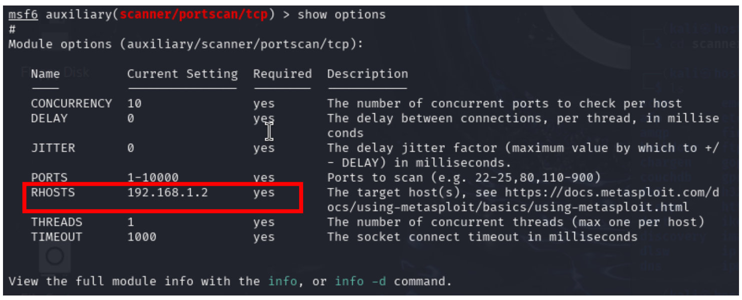


```
msf6 > use auxiliary/scanner/portscan/tcp
msf6 auxiliary(scanner/portscan/tcp) > show options

Module options (auxiliary/scanner/portscan/tcp):

   Name         Current Setting  Required  Description
   ----         ---------------  --------  -----------
   CONCURRENCY  10               yes       The number of concurrent ports to check per host
   DELAY        0                yes       The delay between connections, per thread, in millise
                                           conds
   JITTER       0                yes       The delay jitter factor (maximum value by which to +/
                                           - DELAY) in milliseconds.
   PORTS        1-10000          yes       Ports to scan (e.g. 22-25,80,110-900)
   RHOSTS                        yes       The target host(s), see https://docs.metasploit.com/d
                                           ocs/using-metasploit/basics/using-metasploit.html
   THREADS      1                yes       The number of concurrent threads (max one per host)
   TIMEOUT      1000             yes       The socket connect timeout in milliseconds

View the full module info with the info, or info -d command.
```

Tcp port scanner will show the **mandatory module** in the **name** section.

If you need to set any mandatory modifications needed to run the module. Like targets using RHOSTS, THREADS to 5, or PORTS range.

- set RHOSTS 192.168.1.2
  check again – show options



```
msf6 auxiliary(scanner/portscan/tcp) > show options
#
Module options (auxiliary/scanner/portscan/tcp):

   Name         Current Setting  Required  Description
   ----         ---------------  --------  -----------
   CONCURRENCY  10               yes       The number of concurrent ports to check per host
   DELAY        0                yes       The delay between connections, per thread, in millise
                                           conds
   JITTER       0                yes       The delay jitter factor (maximum value by which to +/
                                           - DELAY) in milliseconds.
   PORTS        1-10000          yes       Ports to scan (e.g. 22-25,80,110-900)
   RHOSTS       192.168.1.2      yes       The target host(s), see https://docs.metasploit.com/d
                                           ocs/using-metasploit/basics/using-metasploit.html
   THREADS      1                yes       The number of concurrent threads (max one per host)
   TIMEOUT      1000             yes       The socket connect timeout in milliseconds

View the full module info with the info, or info -d command.
```

NOW, In the second terminal "monitor scanning network traffic", MSF is generating.

- **Sudo tcpdump –nvi labbr0**

And then, In first terminal run the auxiliary script module from msfconsole, "run" in the msfconsole shell.

Check "**services**" and info is blank.

```
host           port  proto  name    state  info
----           ----  -----  ----    -----  ----
192.168.1.254  22    tcp    ssh     open
192.168.1.254  53    tcp    domain  open
```

Question 7 - Target Enumeration | Service Fingerprtinting and Vulnerability Scanning

Answer 7- As I have not found any details of the service yet. To get the details, let's focus on the (SMB) Service Message Block services and use an msf scanner script to check for the details of the running services.

Steps:

1. **Back**
2. **Use auxiliary/scanner/smb**

```
No     SMB Share Enumeration
  15   auxiliary/scanner/smb/smb_enumusers        .           normal
No     SMB User Enumeration (SAM EnumUsers)
  16   auxiliary/scanner/smb/smb_version          .           normal
No     SMB Version Detection
  17   auxiliary/scanner/smb/smb_uninit_cred      .           normal
Yes    Samba _netr_ServerPasswordSet Uninitialized Credential State
  18   auxiliary/scanner/smb/impacket/wmiexec     2018-03-19  normal
No     WMI Exec
```

Now, use SMB version fingerprinting script to fingerprint versions of SMB services and also host OS versions via service fingerprinitng information.

Use auxiliary/scanner/smb/smb_version

Question 8- Vulnerability Scanning

Question 9- Bind vs Reserve Payload Types